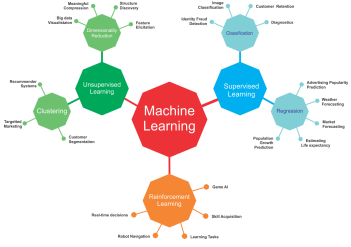
Introduction to Machine Learning

ML-Basics
What is Machine Learning?



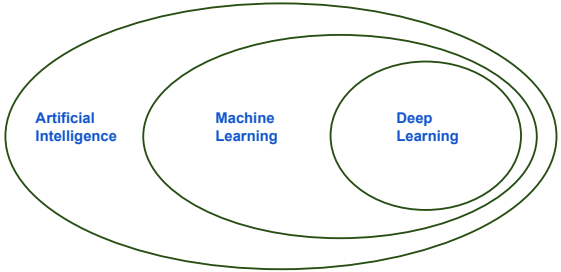
Learning goals

- Understand basic terminology of and connections between ML, AI, DL and statistics
- Know the main directions of ML: Supervised, Unsupervised and Reinforcement Learning

ML IS CHANGING OUR WORLD

- Search engines learn your search preferences
- Recommender systems learn your taste in books, music, movies,...
- Algorithms do automatic stock trading
- Tools can accurately translate between many different languages
- DeepMind beats humans at Go
- Physicians are supported by personalized medicine
- LLMs pretty much revolutionize everything (currently especially coding)
- Data-driven discoveries are made in physics, biology, genetics, astronomy, chemistry, neurology,...
- ...

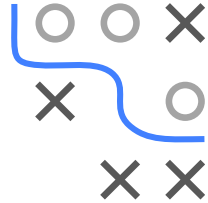
AI, ML AND DL



Many people are confused what these terms actually mean.

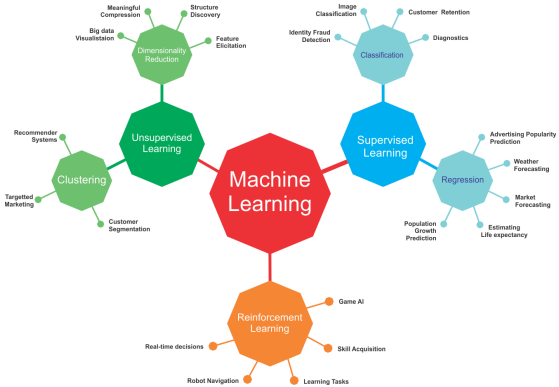
And what does all this have to do with statistics?

ARTIFICIAL INTELLIGENCE



- General term for very large and rapidly developing field.
- No strict definition, but often used when machines perform tasks that could only be solved by humans or are very difficult and assumed to require “intelligence”.
- Started in 1940s - when the computer was invented. Turing and von Neumann immediately asked: If we can formalize computation, can we use that to formalize “thinking”?
- AI includes ML, NLP, computer vision, robotics, planning, search, game playing, intelligent agents, ...
- Nowadays, AI is a “hype” term many people use when they rather mean: ML or ... basic data analysis.

MACHINE LEARNING



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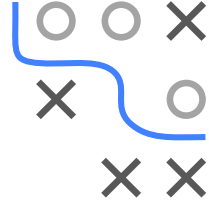
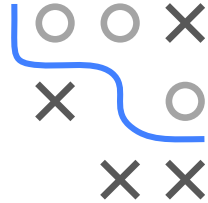


- Mathematically well-defined and solves reasonably narrow tasks.
- Usually construct predictive models from data, instead of explicitly programming them.
- “A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.”
  Tom Mitchell, Carnegie Mellon University, 1998

DEEP LEARNING

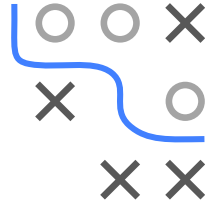


- Subfield of ML which studies neural networks.
- Artificial neural networks are roughly inspired by the human brain, but we treat them as useful, mathematical models.
- ANNs studied for decades (starting in the 1940/50s). DL uses more layers, specific neurons were, e.g., invented for images and computational improvements allow training on large data.
- Can be used on tabular data, but typical applications are images, texts or signals.
- The last 15-20 years have produced remarkable results and imitations of human ability, where the result looked intelligent.

"Any sufficiently advanced technology is indistinguishable from magic."

Arthur C. Clarke's 3rd law

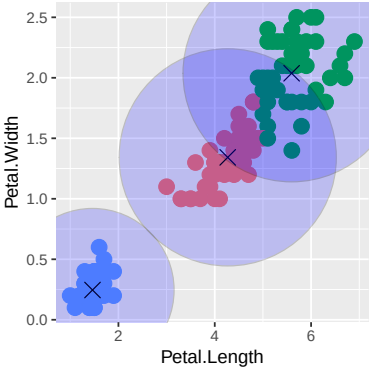
ML VS. STATS



- Historically developed as different fields, but many methods and concepts are pretty much the same.
- ML: Rather accurate predictions with more complex models
- Stats: More interpreting relationships and sound inference.
- Now: Both basically work on same problems with same tools.
- Communities are still divided.
- Often different terminology for the same concepts.
- Most parts of ML we could also call:
  Nonparametric statistics plus efficient numerical optimization.

UNSUPERVISED LEARNING

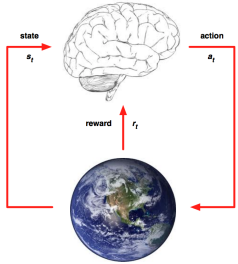
- Data without labels y
- Search for patterns within the inputs $\mathbf{x}$
- *Unsupervised* as there is no “true” output we can optimize against



- Dimensionality reduction (PCA, autoencoders ...); compress information in $\mathcal{X}$
- Clustering: group similar observations
- Outlier detection, anomaly detection
- Association rules

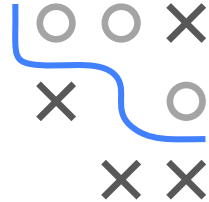
REINFORCEMENT LEARNING

General-purpose framework. At each time step an *agent* interacts with an *environment*. It: observes state; receives reward; executes action.



- Goal: Select actions to maximize future reward.
- Reward signals may be sparse, noisy and delayed.

WHAT COMES NEXT



- **Supervised learning** for regression and classification: predict labels y through features $\mathbf{x}$, based on training data.
- First, we will go through fundamental concepts in supervised ML:
  - What kind of “data” do we learn from?
  - What is a “prediction model”?
  - How can we quantify “predictive performance”?
  - What is a “learning algorithm”
  - How can we operationalize learning?
- We will also introduce first concrete learning algorithms: Linear models, trees and forests.
- More complex stuff comes later.